

The Algebraic Balance Sheet of a Three-Nation Financial System

Symmetric Invariants, Cubic Closure, and Political-Financial Equilibrium

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Abstract

This paper develops an algebraic and financial interpretation of the three-nation financial architecture developed in the preceding paper, *The Three-Nation, Nine-Party Political-Financial Equilibrium*. The central observation is that three national central-bank interest rates r_1, r_2, r_3 generate three elementary symmetric financial accounts,

$$r_A = r_1 + r_2 + r_3,$$

$$r_B = r_1 r_2 + r_2 r_3 + r_3 r_1,$$

and

$$r_C = r_1 r_2 r_3.$$

These quantities determine the cubic

$$r^3 - r_A r^2 + r_B r - r_C = 0,$$

whose roots are precisely the three central-bank rates.

The paper interprets this construction as an algebraic closure condition analogous, at the level of architecture rather than literal accounting identity, to the fundamental balance-sheet relation

$$\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}.$$

The distinction between an accounting identity and an economic equilibrium is emphasized throughout.

The historical development of cubic and quartic algebra by Tartaglia, Cardano, and Ferrari is used to motivate the mathematical structure. Vieta's relations explain why the three aggregate accounts contain exactly the symmetric information necessary to reconstruct the unordered set of three rates. The discriminant characterizes degeneracy and symmetry of the financial roots. The resulting framework also connects naturally to polynomial algebra, group symmetry, network theory, welfare economics, international finance, political economy, game theory, statistical estimation, machine learning, and multi-agent reinforcement learning.

A principal structural result is the Three-Nation Financial Closure Theorem: if the financial architecture requires three genuine, nondegenerate elementary symmetric invariants and a cubic whose roots are the three central-bank rates, then at least three genuine financial agents are required. A two-nation system naturally closes at degree two, while a three-nation system naturally closes at degree three. Thus the cubic is not an arbitrary mathematical decoration; it is the minimal polynomial closure of the three-rate architecture.

The paper further shows how this financial layer nests inside the previously developed three-nation, nine-party political-financial equilibrium. The result is a multilayer architecture in which accounting closure, algebraic closure, monetary policy, political competition, international relations, and strategic equilibrium are distinct but interacting layers.

The paper ends with "The End"

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1 Introduction

The fundamental accounting identity is

$$\boxed{\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}.}$$

This identity expresses a closure condition.

A balance sheet must balance. The resources represented by the asset side must equal the claims represented by liabilities and shareholders' equity.

The present paper develops an analogous closure principle for a three-nation financial system.

Let the three central banks have interest rates

$$r_1, \quad r_2, \quad r_3.$$

Define three common financial accounts:

$$r_A = r_1 + r_2 + r_3,$$

$$r_B = r_1 r_2 + r_2 r_3 + r_3 r_1,$$

and

$$r_C = r_1 r_2 r_3.$$

These quantities generate

$$\boxed{r^3 - r_A r^2 + r_B r - r_C = 0.}$$

By direct factorization,

$$(r - r_1)(r - r_2)(r - r_3) = r^3 - r_A r^2 + r_B r - r_C.$$

Hence the three central-bank rates are the roots of a common cubic.

The resulting architecture can be summarized as

$$\boxed{(r_1, r_2, r_3) \longrightarrow (r_A, r_B, r_C) \longrightarrow P(r) \longrightarrow \{r_1, r_2, r_3\}.}$$

The construction is therefore reversible at the level of the unordered root set.

This paper interprets the cubic as an *algebraic balance sheet* of the three-rate system. The analogy must be stated carefully: the cubic identity is not literally an accounting identity. Rather, both structures perform a closure function. Accounting closes the balance sheet; the polynomial closes the algebraic representation of the financial state.

2 Relationship to the Previous Political-Financial Architecture

The preceding paper established a three-nation, nine-party political-financial equilibrium.[1]

The nations were represented as

$$\mathcal{N} = \{N_1, N_2, N_3\},$$

and each nation contained three political parties:

$$\mathcal{P}_i = \{C_i, L_i, R_i\}.$$

Therefore,

$$|\mathcal{P}| = 3 \times 3 = 9.$$

The political equilibrium could take the differentiated form

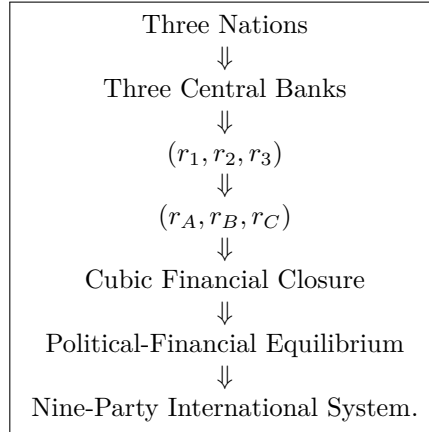
$$(C_1, L_2, R_3),$$

or one of the other permutations of the three political families.

The present paper adds a more primitive financial layer:

$$\text{three nations} \longrightarrow \text{three central banks} \longrightarrow \text{three rates} \longrightarrow \text{three symmetric accounts}.$$

The complete architecture can therefore be represented as



The preceding political-financial paper is therefore not replaced. It is extended downward into the algebraic and accounting structure of the financial system.

3 The Balance-Sheet Principle

A balance sheet has the canonical form

$$A = L + E,$$

where

A = Assets,

L = Liabilities,

E = Shareholders' Equity.

Equivalently,

$$A - L - E = 0.$$

The important mathematical feature is not the particular economic meaning of the three terms but the fact that the system satisfies a closure equation.

The balance-sheet state is therefore constrained to the surface

$$\mathcal{B} = \{(A, L, E) : A - L - E = 0\}.$$

The three-rate financial system has an analogous algebraic constraint:

$$P(r) = r^3 - r_A r^2 + r_B r - r_C = 0.$$

The root r_i belongs to the algebraic state space

$$\mathcal{R} = \{r : P(r) = 0\}.$$

Thus the two systems have the structural form

financial state + closure condition = consistent representation.
--

4 Accounting Identity versus Economic Equilibrium

It is essential to distinguish three different concepts.

4.1 Accounting Identity

The identity

$$A = L + E$$

must hold for a correctly constructed balance sheet.

It is an identity.

4.2 Algebraic Identity

The definitions

$$r_A = r_1 + r_2 + r_3,$$

$$r_B = r_1 r_2 + r_2 r_3 + r_3 r_1,$$

and

$$r_C = r_1 r_2 r_3$$

imply

$$(r - r_1)(r - r_2)(r - r_3) = r^3 - r_A r^2 + r_B r - r_C.$$

This is also an identity.

4.3 Economic Equilibrium

An economic equilibrium requires behavioral conditions.

For example,

$$r_i^* = \arg \max_{r_i} U_i(r_i, r_j, r_k)$$

subject to the relevant institutional and financial constraints.

Thus

identity $\neq$ equilibrium.

The accounting identity guarantees consistency of representation. The algebraic identity guarantees consistency of the polynomial representation. Neither alone proves that the underlying economic choices are optimal or stable.

5 Three Central Banks and Three Rates

Let

$$\mathbf{r} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}.$$

The three central banks are

$$\mathcal{CB}_1, \quad \mathcal{CB}_2, \quad \mathcal{CB}_3.$$

Their policy vector is

$$\mathbf{r} = (r_1, r_2, r_3).$$

The common financial accounts are

$$r_A = e_1(\mathbf{r}),$$

$$r_B = e_2(\mathbf{r}),$$

and

$$r_C = e_3(\mathbf{r}),$$

where e_k denotes the k th elementary symmetric polynomial.

The construction can be depicted as follows.

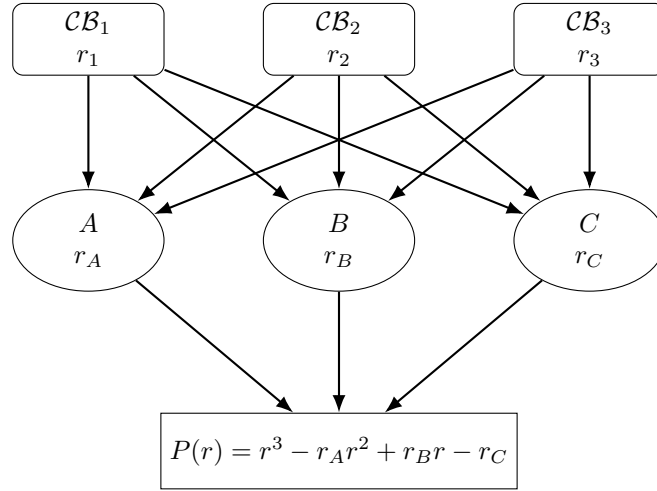


Figure 1: Three central banks, three symmetric accounts, and one common cubic closure equation.

6 Elementary Symmetric Financial Accounts

The three quantities are

$$e_1 = r_1 + r_2 + r_3,$$

$$e_2 = r_1 r_2 + r_2 r_3 + r_3 r_1,$$

and

$$e_3 = r_1 r_2 r_3.$$

We identify

$$r_A = e_1, \quad r_B = e_2, \quad r_C = e_3.$$

These quantities have increasing interaction order.

Account	Expression	Interaction order
A	$r_1 + r_2 + r_3$	First order
B	$r_1 r_2 + r_2 r_3 + r_3 r_1$	Second order
C	$r_1 r_2 r_3$	Third order

The architecture therefore contains

$$\boxed{\text{individual} \rightarrow \text{bilateral} \rightarrow \text{trilateral}}$$

financial information.

7 Vieta's Relations

Consider the monic cubic

$$P(r) = r^3 - ar^2 + br - c.$$

If its roots are

$$r_1, r_2, r_3,$$

then Vieta's relations imply

$$r_1 + r_2 + r_3 = a,$$

$$r_1 r_2 + r_2 r_3 + r_3 r_1 = b,$$

and

$$r_1 r_2 r_3 = c.$$

Therefore, identifying

$$a = r_A, \quad b = r_B, \quad c = r_C,$$

gives

$$\boxed{r^3 - r_A r^2 + r_B r - r_C = (r - r_1)(r - r_2)(r - r_3)}.$$

This is the algebraic core of the architecture.

8 The Cubic as an Algebraic Balance Sheet

The three aggregate accounts

$$(r_A, r_B, r_C)$$

play the role of a compressed state representation.

The polynomial

$$P(r) = r^3 - r_A r^2 + r_B r - r_C$$

is the closure equation.

The roots are the underlying financial states.

Thus:

Aggregate accounts (r_A, r_B, r_C) $\downarrow$ financial summary	$\longrightarrow$ $\longrightarrow$ $\longleftrightarrow$	Closure polynomial $P(r)$ $\downarrow$ $\{r_1, r_2, r_3\}$.
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This motivates the term

algebraic balance sheet.

The term is an analogy: the cubic is not a literal accounting balance sheet.

9 Lossless Symmetric Compression

The map

$$\Phi : (r_1, r_2, r_3) \mapsto (r_A, r_B, r_C)$$

is invariant under permutations of the rates.

For any

$$\sigma \in S_3,$$

we have

$$\Phi(r_1, r_2, r_3) = \Phi(r_{\sigma(1)}, r_{\sigma(2)}, r_{\sigma(3)}).$$

Therefore the accounts do not identify the labels of the nations.

They identify the unordered multiset

$$\{r_1, r_2, r_3\}.$$

If the roots are distinct, the polynomial has six possible assignments of the roots to the three nations:

$$3! = 6.$$

Thus algebraic recovery and institutional identification are separate problems.

10 Root Assignment

Suppose

$$r_1 > r_2 > r_3.$$

An institutional convention may define

$$N_1 = \arg \max_i r_i,$$

$$N_2 = \arg \text{median}_i r_i,$$

and

$$N_3 = \arg \min_i r_i.$$

Then the ordering convention restores the labels.

More generally, let

$$\pi : \{r_1, r_2, r_3\} \rightarrow \{N_1, N_2, N_3\}$$

be the institutional root-assignment rule.

The complete financial architecture is therefore

symmetric recovery + institutional labeling.

11 Simultaneous Central-Bank Solution

All three central banks can solve

$$P(r) = 0$$

using the common accounts.

Let

$$\mathcal{R} = \{r : P(r) = 0\}.$$

Then

$$\mathcal{R} = \{r_1, r_2, r_3\}$$

when the rates are real.

The distributed architecture is

$$\boxed{(r_A, r_B, r_C) \longrightarrow P(r) \longrightarrow \mathcal{R}.}$$

Each central bank obtains the same root set.

A separate institutional rule determines which central bank adopts which root.

12 Cardano and the Cubic Financial Equation

The historical importance of Tartaglia and Cardano is that the general cubic became explicitly solvable.

Starting from

$$r^3 - r_A r^2 + r_B r - r_C = 0,$$

make the substitution

$$r = x + \frac{r_A}{3}.$$

The quadratic term disappears, producing

$$x^3 + px + q = 0,$$

where

$$p = r_B - \frac{r_A^2}{3},$$

and

$$q = -\frac{2r_A^3}{27} + \frac{r_A r_B}{3} - r_C.$$

Cardano's formula then gives

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}}.$$

Consequently,

$$\boxed{r = \frac{r_A}{3} + \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}},}$$

where

$$\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3.$$

The remaining roots can be obtained by multiplying the cube-root terms by the complex cube roots of unity.

13 Tartaglia, Cardano, and Ferrari

The historical development is directly relevant to the present architecture.

Scipione del Ferro discovered a method for a class of cubic equations but did not initially publish it.

Tartaglia independently developed a cubic solution and famously defeated Antonio Maria Fiore in a public mathematical contest.

Cardano obtained Tartaglia's method and published it in *Ars Magna* in 1545, together with Ferrari's solution of the quartic.

Ferrari's contribution was the general solution of the quartic, reducing the quartic to an auxiliary cubic and then to quadratic equations.

The historical chain is therefore

$$\boxed{\text{del Ferro} \rightarrow \text{Tartaglia} \rightarrow \text{Cardano} \rightarrow \text{Ferrari.}}$$

The mathematical chain is

$$\boxed{\text{quadratic} \rightarrow \text{cubic} \rightarrow \text{quartic} \rightarrow \text{higher-degree algebra.}}$$

The cubic is consequently one of the historical foundations of modern algebra.

14 Why Ferrari Matters

Ferrari demonstrated that a quartic could be reduced to an auxiliary cubic.

Schematically,

$$\boxed{\text{quartic} \rightarrow \text{resolvent cubic} \rightarrow \text{quadratic equations.}}$$

This illustrates a general mathematical principle:

$$\boxed{\text{complex problem} \rightarrow \text{structured auxiliary problem} \rightarrow \text{solution.}}$$

The three-nation financial architecture uses a related idea:

$$\boxed{\text{three-rate system} \rightarrow \text{symmetric invariants} \rightarrow \text{common cubic} \rightarrow \text{rate recovery.}}$$

The financial polynomial is therefore a compact computational representation of the underlying multi-agent system.

15 The Discriminant and Financial Degeneracy

The Cardano discriminant is

$$\Delta = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3.$$

For the depressed cubic:

- $\Delta > 0$ gives one real root and two complex conjugate roots;
- $\Delta = 0$ gives a repeated root;
- $\Delta < 0$ gives three distinct real roots.

For the three-nation financial interpretation, three distinct real rates require

$$\boxed{\Delta < 0.}$$

A repeated financial rate corresponds to

$$\Delta = 0.$$

Thus the discriminant acts as a degeneracy diagnostic.

16 Symmetric Financial Equilibrium

If

$$r_1 = r_2 = r_3 = r^*,$$

then

$$r_A = 3r^*,$$

$$r_B = 3(r^*)^2,$$

and

$$r_C = (r^*)^3.$$

The polynomial becomes

$$r^3 - 3r^*r^2 + 3(r^*)^2r - (r^*)^3 = 0,$$

or

$$(r - r^*)^3 = 0.$$

Thus complete financial symmetry corresponds to a triple root.

This provides an algebraic representation of the symmetric equilibrium.

17 Symmetry Breaking

Suppose the rates separate:

$$r_1 \neq r_2 \neq r_3.$$

Then the cubic has distinct roots and

$$\Delta < 0$$

for three real rates.

The financial symmetry is broken.

Hence

$$\text{symmetry} \leftrightarrow \text{repeated roots,}$$

while

$$\text{financial differentiation} \leftrightarrow \text{distinct roots.}$$

This parallels the political symmetry-breaking analysis of the preceding nine-party equilibrium.

18 The Minimum Number of Nations

We now state the principal structural theorem.

18.1 Theorem 1: Three-Nation Financial Closure Theorem

Suppose a financial architecture satisfies:

1. each nation has one genuine central-bank rate;
2. the system defines the first three elementary symmetric invariants;
3. the third invariant is required to be nondegenerate;
4. the common characteristic polynomial has those three rates as its roots.

Then the architecture requires at least three nations.

Proof

With one nation,

$$e_1 = r_1$$

is the only nontrivial elementary symmetric invariant.

With two nations,

$$e_1 = r_1 + r_2$$

and

$$e_2 = r_1 r_2$$

are the highest-order invariants.

A genuine third-order invariant

$$e_3 = r_1 r_2 r_3$$

requires a third rate.

Therefore a system requiring

$$r_C = e_3 \neq 0$$

cannot be generated by fewer than three genuine central banks.

Hence

$$\boxed{N_{\min} = 3}.$$

□

19 Why Two Nations Produce a Quadratic

With two central-bank rates,

$$r_1, r_2,$$

define

$$r_A = r_1 + r_2$$

and

$$r_B = r_1 r_2.$$

Then

$$(r - r_1)(r - r_2) = r^2 - r_A r + r_B.$$

Therefore the natural characteristic equation is

$$\boxed{r^2 - r_A r + r_B = 0.}$$

There is no genuine third-order invariant.

The progression is

N	Elementary invariants	Characteristic polynomial
1	e_1	linear
2	e_1, e_2	quadratic
3	e_1, e_2, e_3	cubic
4	e_1, e_2, e_3, e_4	quartic

Thus polynomial degree and number of financial agents coincide under this construction.

20 Degenerate Two-Nation Embedding

There is a mathematical loophole.

Suppose one artificially sets

$$r_3 = 0.$$

Then

$$r_C = r_1 r_2 r_3 = 0.$$

The cubic becomes

$$r^3 - r_A r^2 + r_B r = 0,$$

so

$$r(r^2 - r_A r + r_B) = 0.$$

The roots are

$$0, r_1, r_2.$$

Mathematically this is a cubic.

Economically, however, the third root represents an artificial third state with

$$r_3 = 0.$$

It is therefore not a genuinely two-nation implementation of a nondegenerate three-rate architecture.

The theorem should consequently be stated as an architectural result:

nondegenerate three-rate closure $\Rightarrow$ at least three genuine nations.
--

21 Generalization to N Nations

Let there be N rates

$$r_1, \dots, r_N.$$

Define

$$e_k = \sum_{1 \leq i_1 < \dots < i_k \leq N} r_{i_1} \cdots r_{i_k}.$$

Then

$$\prod_{i=1}^N (r - r_i) = r^N - e_1 r^{N-1} + e_2 r^{N-2} - \dots + (-1)^N e_N.$$

Thus

N rates $\longleftrightarrow N$ elementary symmetric invariants.
--

The three-nation architecture is therefore the $N = 3$ member of a general polynomial financial architecture.

22 Graph-Theoretic Interpretation

Let the nations be vertices of a complete graph.

For three nations,

$$K_3$$

has three vertices and three edges.

The pairwise account is

$$r_B = r_1 r_2 + r_2 r_3 + r_3 r_1.$$

Each term corresponds to one edge:

$$(1, 2), \quad (2, 3), \quad (3, 1).$$

The product

$$r_C = r_1 r_2 r_3$$

corresponds to the complete three-vertex interaction.

The architecture can therefore be interpreted as

vertices + edges + three-way interaction.

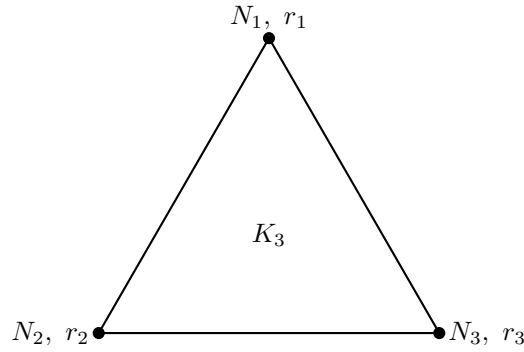


Figure 2: Three nations as a complete financial interaction graph.

23 Physics Analogy

The polynomial representation is analogous to the use of characteristic polynomials in linear algebra and physics.

For a matrix M , its eigenvalues satisfy

$$\det(\lambda I - M) = 0.$$

The characteristic polynomial has coefficients that are invariant under similarity transformations.

Similarly, the financial polynomial

$$P(r) = \prod_{i=1}^3 (r - r_i)$$

has coefficients invariant under permutations of the three rates.

Thus:

eigenvalues $\leftrightarrow$ financial rates,

and

characteristic coefficients $\leftrightarrow$ symmetric financial accounts.

This is a mathematical analogy rather than a claim that central-bank rates are literally eigenvalues of a physical Hamiltonian.

24 Chemical Analogy

Chemical reaction systems often involve first-, second-, and higher-order interaction terms.

The financial construction has an analogous interaction hierarchy:

$$e_1 \rightarrow e_2 \rightarrow e_3.$$

Thus

$$r_A$$

represents first-order aggregation,

$$r_B$$

represents pairwise interaction,
and

$$r_C$$

represents three-way interaction.

Again, this is a structural analogy rather than a chemical identity.

25 Biological Analogy

Biological systems often distinguish between individual effects, pairwise interactions, and higher-order interactions.

The financial system can similarly be represented as

$$\mathcal{F} = \mathcal{F}_1 + \mathcal{F}_2 + \mathcal{F}_3,$$

where

$$\mathcal{F}_1 \sim e_1,$$

$$\mathcal{F}_2 \sim e_2,$$

$$\mathcal{F}_3 \sim e_3.$$

The third-order term captures information unavailable in a purely two-agent model.

26 Political Economy

The financial rates interact with the political equilibrium.

Let the governing party in nation i be

$$p_i \in \{C_i, L_i, R_i\}.$$

Then the financial policy function can be written

$$r_i = R_i(p_i, X_i, \varepsilon_i),$$

where X_i contains economic and institutional variables.

Conversely, political attractiveness depends on the financial state:

$$V_{ip} = V_{ip}(r_i, X_i).$$

Therefore,

$$p_i \rightarrow r_i \rightarrow V_{ip} \rightarrow p_i$$

creates a political-financial feedback loop.

27 The Joint Fixed Point

Let

$$\mathbf{p} = (p_1, p_2, p_3)$$

and

$$\mathbf{r} = (r_1, r_2, r_3).$$

Let

$$\Phi : \mathbf{r} \mapsto \mathbf{p}$$

be the electoral-response operator, and

$$\Psi : \mathbf{p} \mapsto \mathbf{r}$$

be the policy-to-financial operator.
Then equilibrium requires

$$\mathbf{p}^* = \Phi(\mathbf{r}^*)$$

and

$$\mathbf{r}^* = \Psi(\mathbf{p}^*).$$

Hence

$$\mathbf{r}^* = (\Psi \circ \Phi)(\mathbf{r}^*).$$

The cubic closure is then an additional algebraic constraint:

$$P_{\mathbf{r}^*}(r) = 0.$$

Thus the complete equilibrium satisfies

$$\begin{cases} \mathbf{p}^* = \Phi(\mathbf{r}^*), \\ \mathbf{r}^* = \Psi(\mathbf{p}^*), \\ P_{\mathbf{r}^*}(r) = 0. \end{cases}$$

28 International Finance

The three rates determine bilateral interest-rate spreads:

$$\Delta_{12} = r_1 - r_2,$$

$$\Delta_{23} = r_2 - r_3,$$

and

$$\Delta_{31} = r_3 - r_1.$$

Under covered interest parity, exchange-rate forward relationships can depend on these differentials.
A stylized relation is

$$\frac{F_{ij}}{S_{ij}} = \frac{1 + r_i}{1 + r_j}.$$

Thus the three-rate architecture creates a triangular international financial network.

The rates are not isolated national variables; they influence capital flows, exchange rates, reserves, and financial arbitrage.

29 Arbitrage

Suppose

$$r_1 > r_2.$$

Ignoring risk and transaction costs, capital has an incentive to move toward the higher return.

Let

$$K_{12}$$

denote the total friction between nations 1 and 2.

A simplified net arbitrage condition is

$$r_1 - r_2 > K_{12}.$$

In a fully integrated system, these flows feed back into the rates.

Therefore

$$r_i \rightarrow \text{capital flow} \rightarrow \text{financial conditions} \rightarrow r_i.$$

This supplies an economic mechanism for the fixed-point operator Ψ .

30 Welfare Economics

Let national welfare be

$$W_i = Y_i + \omega_L L_i + \omega_F F_i + \omega_S S_i - K_i.$$

Aggregate welfare is

$$W = W_1 + W_2 + W_3.$$

The financial architecture may create positive externalities through financial integration, but also negative externalities through contagion.

Let

$$X_{ij} = \frac{\partial W_i}{\partial r_j}.$$

Then the welfare externality matrix is

$$X = \begin{pmatrix} \frac{\partial W_1}{\partial r_1} & \frac{\partial W_1}{\partial r_2} & \frac{\partial W_1}{\partial r_3} \\ \frac{\partial W_2}{\partial r_1} & \frac{\partial W_2}{\partial r_2} & \frac{\partial W_2}{\partial r_3} \\ \frac{\partial W_3}{\partial r_1} & \frac{\partial W_3}{\partial r_2} & \frac{\partial W_3}{\partial r_3} \end{pmatrix}.$$

This matrix connects monetary policy to international welfare.

31 Game Theory

Each central bank may be represented as a player.

Let

$$U_i(r_i, r_j, r_k)$$

be central-bank objective i .

A Nash equilibrium satisfies

$$r_i^* \in \arg \max_{r_i} U_i(r_i, r_j^*, r_k^*)$$

for each i .

The polynomial closure then imposes

$$(r_1^*, r_2^*, r_3^*) \in \mathcal{R},$$

where $\mathcal{R}$ is the set of root triples generated by the common accounts.

Thus the model combines behavioral equilibrium and algebraic consistency.

32 Strategic Policy Coordination

Suppose the three central banks coordinate on the common account vector

$$\mathbf{a} = (r_A, r_B, r_C).$$

Then they solve

$$P(r; \mathbf{a}) = 0.$$

The common polynomial becomes an information-sharing mechanism.

The coordination architecture is

shared invariants $\rightarrow$ shared polynomial $\rightarrow$ consistent root set.
--

This does not imply that all central banks must choose the same rate.

Quite the opposite: the common polynomial can encode three distinct rates.

33 Diplomacy and International Relations

The three-nation structure creates three bilateral relationships:

$$G_{12}, \quad G_{23}, \quad G_{31}.$$

Each may contain financial, trade, diplomatic, technological, and security components.

Define

$$G_{ij} = (F_{ij}, T_{ij}, D_{ij}, M_{ij}, X_{ij}).$$

The complete international system is therefore a multilayer network.

The financial polynomial provides a common algebraic layer over this network.

34 Warfare Economics

Security expenditure may be represented by

$$S_i.$$

Expected warfare cost is

$$C_W = C_{\text{military}} + C_{\text{destruction}} + C_{\text{escalation}} + C_{\text{retaliation}}.$$

Political-financial conditions influence the fiscal capacity to sustain security expenditure.
Thus

$$r_i \rightarrow \text{fiscal capacity} \rightarrow S_i \rightarrow \text{deterrence} \rightarrow G_{ij}^M.$$

The financial architecture therefore cannot be completely separated from strategic relations.

35 Psychology and Behavioral Economics

Central-bank and voter decisions may depart from the frictionless rational benchmark.

Let an agent assign subjective probability

$$\hat{P}(E)$$

rather than objective probability

$$P(E).$$

Then policy decisions may depend upon

$$E_{\hat{P}}[U]$$

rather than

$$E_P[U].$$

Behavioral factors can therefore influence the mapping

$$\Psi : \mathbf{p} \rightarrow \mathbf{r}.$$

Political psychology similarly influences

$$\Phi : \mathbf{r} \rightarrow \mathbf{p}.$$

No psychiatric classification is implied by such behavioral modeling.

36 Statistics and Econometrics

Suppose observed rates are

$$r_{it} = \mu_i + \beta X_{it} + \varepsilon_{it}.$$

The aggregate account is

$$r_{A,t} = r_{1t} + r_{2t} + r_{3t}.$$

The pairwise account is

$$r_{B,t} = r_{1t}r_{2t} + r_{2t}r_{3t} + r_{3t}r_{1t},$$

and the third-order account is

$$r_{C,t} = r_{1t}r_{2t}r_{3t}.$$

The resulting time series

$$(r_{A,t}, r_{B,t}, r_{C,t})$$

can be analyzed directly.

A useful empirical question is whether the three-account representation contains enough information to reconstruct the observed rate multiset.

The exact algebraic answer is yes, subject to numerical precision and root identification.

37 Numerical Stability

Although the algebraic mapping is exact symbolically, numerical recovery of roots can be sensitive when roots are nearly repeated.

Suppose

$$r_1 \approx r_2.$$

Then

$$\Delta \approx 0.$$

Small errors in the coefficients can produce comparatively large errors in the recovered roots. Thus the condition number of the inverse mapping

$$(r_A, r_B, r_C) \rightarrow (r_1, r_2, r_3)$$

may become large near discriminant zero.

This is an important distinction between:

exact algebra

and

finite-precision computation.

38 Computer Science Interpretation

The common accounts form a compressed representation of the financial state.

Let

$$\text{Encode}(r_1, r_2, r_3) = (r_A, r_B, r_C).$$

Let

$$\text{Decode}(r_A, r_B, r_C) = \{r_1, r_2, r_3\}.$$

Then, ideally,

$$\text{Decode} \circ \text{Encode} = \text{Identity}$$

on unordered triples.

This is a lossless encoding modulo permutation.

The representation is therefore analogous to a canonical invariant-based data structure.

39 Algorithm for Financial State Recovery

Pseudocode

```

INPUT:
    r_A, r_B, r_C

CONSTRUCT:
    P(r) = r^3 - r_A*r^2 + r_B*r - r_C

COMPUTE:
    p = r_B - r_A^2/3

    q = -2*r_A^3/27
        + r_A*r_B/3
        - r_C

    Delta = (q/2)^2 + (p/3)^3

IF Delta > 0:
    compute one real root
    compute complex conjugate pair

ELSE IF Delta = 0:
    compute repeated roots

ELSE:
    compute three distinct real roots

RETURN:
    unordered root set {r_1,r_2,r_3}

APPLY:
    institutional root-labeling rule

RETURN:
    labeled rates assigned to N_1, N_2, N_3

```

40 Machine Learning Representation

The financial state can be represented by

$$X = (r_A, r_B, r_C).$$

A machine-learning model may estimate

$$\hat{\mathbf{r}} = f_{\theta}(X, Z),$$

where Z contains additional economic information.

However, the exact polynomial structure gives a hard algebraic constraint:

$$P_{\hat{\mathbf{r}}}(\hat{r}_i) = 0.$$

A learning system can therefore incorporate the algebraic structure into its loss function. For example,

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{algebra}},$$

where

$$\mathcal{L}_{\text{algebra}} = \sum_{i=1}^3 \left| \hat{r}_i^3 - r_A \hat{r}_i^2 + r_B \hat{r}_i - r_C \right|^2.$$

This is an example of a physics-informed or structure-informed learning principle generalized to financial algebra.

41 Data Science

A three-rate financial dataset may be represented as

$$D = \{t, r_{1t}, r_{2t}, r_{3t}\}_{t=1}^T.$$

Transform it into

$$D' = \{t, r_{A,t}, r_{B,t}, r_{C,t}\}_{t=1}^T.$$

The transformation is deterministic:

$$D \rightarrow D'.$$

The inverse reconstruction is obtained by solving the cubic independently at each t . This produces a useful empirical test:

$$\hat{r}_{it} \stackrel{?}{=} r_{it}.$$

The reconstruction error is

$$E_t = \min_{\pi \in S_3} \sum_{i=1}^3 \left(\hat{r}_{\pi(i),t} - r_{i,t} \right)^2.$$

The permutation minimizes the labeling ambiguity.

42 Multi-Agent Reinforcement Learning

The three central banks can be represented as three agents.

Let

$$a_i$$

be the policy action of central bank i .

The state is

$$x_t = (r_{1t}, r_{2t}, r_{3t}, r_{A,t}, r_{B,t}, r_{C,t}, X_t).$$

The transition equation is

$$x_{t+1} = F(x_t, a_t, \varepsilon_t).$$

Each central bank maximizes

$$J_i = E \left[\sum_{t=0}^{\infty} \beta^t U_i(x_t, a_t) \right].$$

A multi-agent equilibrium requires no profitable unilateral policy deviation. The polynomial constraint can be embedded into the state transition.

43 Information-Theoretic Interpretation

The raw financial state is

$$\mathbf{r} = (r_1, r_2, r_3).$$

The symmetric representation is

$$\mathbf{e} = (e_1, e_2, e_3).$$

The transformation removes labeling information but retains symmetric information. Therefore

$$I(\mathbf{r}; \mathbf{e})$$

is high, while the information concerning the permutation label is absent. The architecture deliberately separates:

economic state

from

institutional identity.

44 Political-Financial Extension

The preceding nine-party architecture contains

$$\mathcal{P} = \{C_1, L_1, R_1, C_2, L_2, R_2, C_3, L_3, R_3\}.$$

The complete state becomes

$$\mathcal{X} = (\mathbf{r}, r_A, r_B, r_C, \mathbf{p}, \mathbf{v}, G, \Pi),$$

where:

- $\mathbf{r}$ = central-bank rates,
- (r_A, r_B, r_C) = symmetric financial accounts,
- $\mathbf{p}$ = governing parties,
- $\mathbf{v}$ = electoral shares,
- G = international network,
- Π = coalition structure.

The political-financial equilibrium satisfies

$$\begin{cases} \mathbf{p}^* = \Phi(\mathbf{r}^*), \\ \mathbf{r}^* = \Psi(\mathbf{p}^*), \\ P_{\mathbf{r}^*}(r) = 0, \\ \Pi^* \text{ is coalition-stable.} \end{cases}$$

This is the complete closure system.

45 A Unified Architecture

The entire system can be represented as

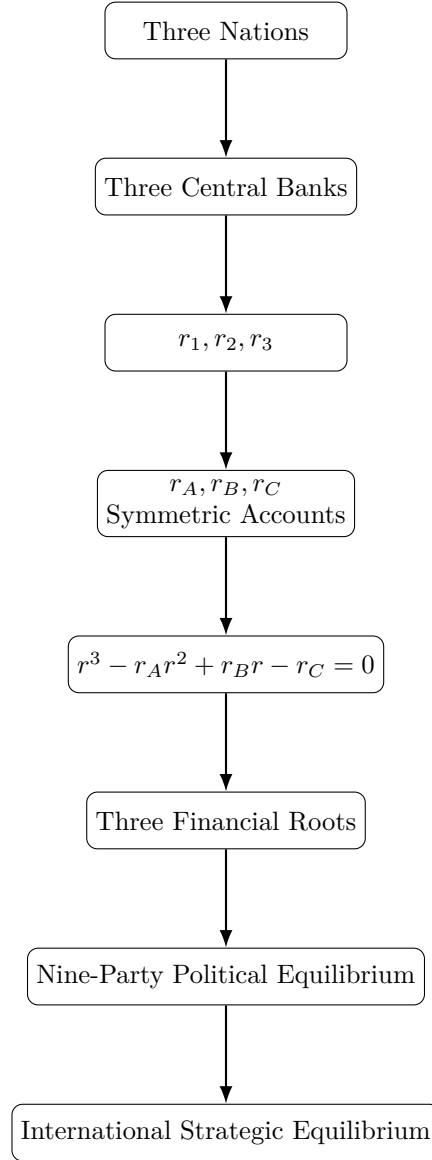


Figure 3: The complete accounting–algebraic–political architecture.

46 The Accounting Interpretation

The analogy can now be stated precisely.

The balance-sheet identity is

$$A - L - E = 0.$$

The cubic identity is

$$r^3 - r_A r^2 + r_B r - r_C = 0$$

when evaluated at any one of the three roots.

In both cases, a structured collection of quantities is constrained by a closure relation.

However,

$$\boxed{A = L + E}$$

is an accounting identity, while

$$\boxed{P(r_i) = 0}$$

is a polynomial identity.

Neither should be confused with an equilibrium condition.

The economic equilibrium requires behavior, incentives, institutions, expectations, and market clearing.

47 A General Closure Principle

The common structure can be summarized as follows.

Let a system contain primitive variables

$$x_1, \dots, x_N.$$

Construct invariants

$$I_1, \dots, I_N.$$

Then define a closure object

$$\mathcal{C}(x; I_1, \dots, I_N) = 0.$$

For the three-rate financial system,

$$x_i = r_i,$$

and

$$(I_1, I_2, I_3) = (r_A, r_B, r_C).$$

The closure object is

$$\mathcal{C}(r) = r^3 - r_A r^2 + r_B r - r_C.$$

Thus the algebraic balance sheet is a special case of a broader invariant-based closure principle.

48 The Three-Nation Lower Bound Revisited

The lower-bound result can now be understood from three independent perspectives.

48.1 Algebraic Perspective

A cubic requires three roots counted with multiplicity.

48.2 Symmetric-Polynomial Perspective

Three elementary symmetric invariants require three primitive variables.

48.3 Network Perspective

The third-order product

$$r_1 r_2 r_3$$

requires three interacting financial agents.

Therefore three independent arguments converge on

$$\boxed{N_{\min} = 3.}$$

49 A Stronger Theorem

Theorem 2 (Three-Layer Financial Closure).

Suppose a financial architecture has:

$$\mathcal{N} = \{N_1, N_2, N_3\},$$

with rates

$$r_1, r_2, r_3,$$

and defines

$$r_A = e_1, \quad r_B = e_2, \quad r_C = e_3.$$

Then:

1. the aggregate representation is invariant under S_3 ;
2. the three rates are roots of a unique monic cubic;
3. three distinct real rates exist exactly when the cubic discriminant is negative;
4. a repeated rate corresponds to a vanishing discriminant;
5. a genuine nonzero third-order invariant requires at least three genuine financial agents.

Proof

Items 1 and 2 follow immediately from the elementary symmetric polynomial representation and Vieta's relations.

Item 3 follows from the standard classification of the depressed cubic by the Cardano discriminant.

Item 4 follows because repeated roots imply a zero polynomial discriminant.

For item 5, e_3 is a product of three distinct primitive rate variables. With fewer than three agents, no nondegenerate third-order elementary symmetric invariant exists.

Therefore all five claims hold.

□

50 Implications for the Three-Nation Architecture

The theorem gives the financial architecture a precise algebraic foundation.

The three nations are not merely a geopolitical choice.

Under the specified construction, they are the minimum number of financial agents required for the complete sequence

$$e_1 \rightarrow e_2 \rightarrow e_3.$$

The sequence corresponds to

$$\text{aggregate} \rightarrow \text{bilateral} \rightarrow \text{trilateral}.$$

The financial system therefore closes exactly at three nations.

51 Relationship to the Nine-Party Equilibrium

The preceding paper supplied three political parties per nation.

Therefore,

$$3 \text{ nations} \times 3 \text{ parties} = 9 \text{ parties}.$$

The present paper supplies three central-bank rates and three symmetric financial accounts.

The complete architecture therefore contains:

3 nations, 3 central banks, 3 rates, 3 financial accounts, 9 parties.

This produces a particularly compact structural hierarchy:

$3 \rightarrow 3 \rightarrow 3 \rightarrow 9.$
--

The first three entries describe the financial system; the final entry describes the political system.

52 A Possible Equilibrium Configuration

Suppose

$$r_1 > r_2 > r_3,$$

and suppose the political equilibrium is

$$(C_1, L_2, R_3).$$

Then the system can be interpreted as having three differentiated financial and political roles:

$$N_1 \rightarrow \text{capital and investment},$$

$$N_2 \rightarrow \text{labor and stabilization},$$

and

$$N_3 \rightarrow \text{security and strategic autonomy}.$$

The financial polynomial remains symmetric:

$$P(r) = (r - r_1)(r - r_2)(r - r_3),$$

while the political assignment is not symmetric.

This produces a useful distinction:

financial algebra can remain symmetric while political institutions break symmetry.

53 International Strategic Equilibrium

Let the three governments choose strategic actions

$$u_1, u_2, u_3.$$

Their utilities are

$$U_i = U_i(r_1, r_2, r_3, u_1, u_2, u_3).$$

The strategic equilibrium satisfies

$$u_i^* \in \arg \max_{u_i} U_i(u_i, u_j^*, u_k^*).$$

The financial polynomial remains an invariant constraint on the rate vector.

Thus the complete equilibrium is

$(\mathbf{r}^*, \mathbf{p}^*, \mathbf{u}^*)$
--

subject to the algebraic closure condition.

54 Policy Robustness

Let the financial state be perturbed:

$$r_i \rightarrow r_i + \varepsilon_i.$$

Then

$$r_A \rightarrow r_A + \varepsilon_1 + \varepsilon_2 + \varepsilon_3.$$

The second invariant changes according to

$$\delta r_B = (r_2 + r_3)\varepsilon_1 + (r_1 + r_3)\varepsilon_2 + (r_1 + r_2)\varepsilon_3 + O(\|\varepsilon\|^2).$$

The third invariant changes according to

$$\delta r_C = r_2 r_3 \varepsilon_1 + r_1 r_3 \varepsilon_2 + r_1 r_2 \varepsilon_3 + O(\|\varepsilon\|^2).$$

Thus perturbations propagate through all three accounts.

55 Jacobian of the Invariant Transformation

The Jacobian of

$$(r_1, r_2, r_3) \mapsto (r_A, r_B, r_C)$$

is

$$J = \begin{pmatrix} 1 & 1 & 1 \\ r_2 + r_3 & r_1 + r_3 & r_1 + r_2 \\ r_2 r_3 & r_1 r_3 & r_1 r_2 \end{pmatrix}.$$

Its determinant is, up to sign,

$$\det J = (r_1 - r_2)(r_1 - r_3)(r_2 - r_3).$$

Therefore,

$$\boxed{\det J \neq 0}$$

when the three rates are distinct.

But if two rates coincide,

$$r_i = r_j,$$

then

$$\det J = 0.$$

This provides a local inverse-function interpretation of the nondegeneracy condition.

Distinct roots imply locally invertible invariant coordinates.

Repeated roots destroy local invertibility.

56 The Vandermonde Structure

The determinant above is the Vandermonde determinant:

$$V = (r_1 - r_2)(r_1 - r_3)(r_2 - r_3).$$

Therefore

$$V^2$$

is proportional to the polynomial discriminant.

The same quantity simultaneously measures:

1. distinctness of the financial rates;
2. local invertibility of the symmetric-coordinate map;
3. algebraic separation of the roots;
4. sensitivity of root recovery.

Thus a single mathematical object connects algebraic, numerical, and financial notions of degeneracy.

57 Computational Complexity

Constructing the three accounts requires constant computational complexity:

$$O(1).$$

For N nations, computing all elementary symmetric polynomials naively can require

$$O(2^N)$$

operations, although dynamic-programming methods reduce this substantially.

For the three-nation case, the computation is trivial.

Solving the cubic by a closed-form method is also constant-time in the formal algebraic model.

Numerically, robust polynomial-root algorithms are preferable to direct radical evaluation near repeated roots.

58 Numerical Root Recovery

In computational applications, one may construct the companion matrix

$$M = \begin{pmatrix} 0 & 0 & r_C \\ 1 & 0 & -r_B \\ 0 & 1 & r_A \end{pmatrix}.$$

Its characteristic polynomial is

$$\det(rI - M) = r^3 - r_A r^2 + r_B r - r_C.$$

Therefore the rates can be recovered as the eigenvalues of M :

$$\boxed{\{r_1, r_2, r_3\} = \text{eig}(M)}.$$

This gives an alternative computational route to Cardano's formula.

It is particularly useful in numerical data analysis.

59 Machine Learning with Algebraic Constraints

Suppose a model predicts

$$\hat{r}_1, \hat{r}_2, \hat{r}_3.$$

Define predicted symmetric invariants

$$\hat{r}_A = \hat{r}_1 + \hat{r}_2 + \hat{r}_3,$$

$$\hat{r}_B = \hat{r}_1 \hat{r}_2 + \hat{r}_2 \hat{r}_3 + \hat{r}_3 \hat{r}_1,$$

and

$$\hat{r}_C = \hat{r}_1 \hat{r}_2 \hat{r}_3.$$

A structure-preserving loss can be

$$\mathcal{L} = \mathcal{L}_{\text{prediction}} + \lambda_A(\widehat{r}_A - r_A)^2 + \lambda_B(\widehat{r}_B - r_B)^2 + \lambda_C(\widehat{r}_C - r_C)^2.$$

This forces the learned rate vector to respect the invariant structure.

60 A World-Model Interpretation

A world model for the three-nation system may contain latent state

$$z_t = (\mathbf{r}_t, \mathbf{p}_t, G_t).$$

The transition is

$$z_{t+1} = F_\theta(z_t, u_t, \varepsilon_t).$$

The polynomial constraint can be imposed as a structural prior:

$$P_t(r_i) = 0$$

for the relevant roots.

Thus the world model separates:

latent economic dynamics

from

algebraic invariants.

This can reduce the dimensionality of the learned state representation.

61 Normative Interpretation

The accounting analogy also has a philosophical implication.

A well-formed institutional system requires consistency between its parts.

In accounting:

$$A = L + E.$$

In the financial algebra:

$$P(r_i) = 0.$$

In political equilibrium:

$$p_i^* = \Phi_i(\mathbf{r}^*).$$

In strategic equilibrium:

$$u_i^* \in \arg \max_{u_i} U_i.$$

These can be viewed as different forms of institutional consistency.

The complete architecture therefore has nested closure conditions:

accounting closure	$\rightarrow$	algebraic closure	$\rightarrow$	economic closure	$\rightarrow$	political closure	$\rightarrow$	strategic closure
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This should be interpreted as a conceptual hierarchy rather than as a claim that all five conditions are mathematically identical.

62 Main Results

The paper establishes the following structural results.

62.1 Result 1: Symmetric Financial Representation

The three rates can be represented exactly by their three elementary symmetric invariants:

$$(r_1, r_2, r_3) \longrightarrow (r_A, r_B, r_C).$$

62.2 Result 2: Cubic Closure

The rates are the roots of

$$r^3 - r_A r^2 + r_B r - r_C = 0.$$

62.3 Result 3: Three-Nation Lower Bound

A genuine nonzero third-order invariant

$$r_C = r_1 r_2 r_3$$

requires at least three genuine financial agents.

Therefore

$$N_{\min} = 3.$$

62.4 Result 4: Symmetry Detection

Repeated financial rates correspond to vanishing discriminant and vanishing Vandermonde determinant.

62.5 Result 5: Political-Financial Integration

The financial state can enter the nine-party political equilibrium through

$$\mathbf{r} \rightarrow \mathbf{p} \rightarrow \mathbf{r}.$$

63 Conclusion

The central result of this paper is that the three-nation financial system admits a remarkably compact algebraic representation.

Starting with

$$r_1, r_2, r_3,$$

we construct

$$r_A = r_1 + r_2 + r_3,$$

$$r_B = r_1 r_2 + r_2 r_3 + r_3 r_1,$$

and

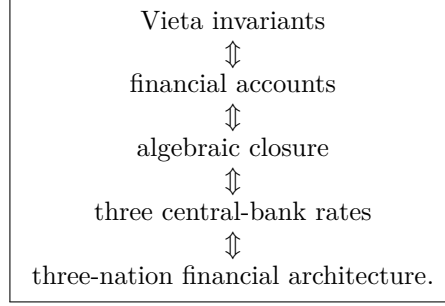
$$r_C = r_1 r_2 r_3.$$

These quantities generate

$$r^3 - r_A r^2 + r_B r - r_C = 0.$$

The cubic is therefore the characteristic equation of the three-rate financial system.

The mathematical structure has several simultaneous interpretations:



The accounting analogy is especially useful when stated correctly.
The identity

$$\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}$$

is an accounting identity. The cubic equation is an algebraic identity. Neither is, by itself, an economic equilibrium.

Nevertheless, both perform the same broad architectural function: they impose a closure condition that prevents the system's components from being internally inconsistent.

The three-nation lower bound follows because the architecture requires the third elementary symmetric invariant

$$e_3 = r_1 r_2 r_3.$$

A two-nation system naturally stops at

$$e_2 = r_1 r_2$$

and therefore closes through a quadratic.

Thus:

1 nation → linear closure, 2 nations → quadratic closure, 3 nations → cubic closure.
--

The historical development from Tartaglia and Cardano to Ferrari supplies the appropriate mathematical context: the cubic became explicitly solvable, the quartic was reduced to an auxiliary cubic, and later Galois theory explained the deeper algebraic structure of polynomial equations.

The present architecture therefore places the three-nation financial system inside a long mathematical tradition:

individual financial states	→ symmetric invariants	→ characteristic polynomial	→ root recovery.
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When combined with the preceding nine-party political-financial model, the result is a multilayer institutional system:

Accounting → Finance → Algebra → Monetary Policy → Politics → International Strategy.

The three-nation architecture is consequently not merely a collection of three states. It is a closed mathematical and institutional system whose financial layer possesses an exact cubic representation and whose political layer contains nine competing parties.

References

- [1] S. Ghosh, *The Three-Nation, Nine-Party Political-Financial Equilibrium*, Kolkata, India, 2026.
- [2] S. Ghosh, *The Three-Nation Financial and Charter Architecture: Symmetric and Hierarchical Equilibria*, Kolkata, India, 2026.
- [3] S. Ghosh, *The Four-Nation Financial Architecture*, Kolkata, India, 2026.
- [4] S. Ghosh, *The Pros and Cons of Charters*, Kolkata, India, 2026.
- [5] G. Cardano, *Ars Magna*, 1545.
- [6] N. Tartaglia, *General references to the algebraic work of Niccolò Tartaglia*, sixteenth century.
- [7] L. Ferrari, *Contribution to the General Solution of the Quartic Equation*, sixteenth century.
- [8] F. Viète, *In artem analyticem isagoge*, 1591.
- [9] R. Bombelli, *L'Algebra*, 1572.
- [10] L. Euler, *Elements of Algebra*, 1770.
- [11] J.-L. Lagrange, *Réflexions sur la résolution algébrique des équations*, 1770.
- [12] P. Ruffini, *Teoria generale delle equazioni*, 1799.
- [13] N. H. Abel, “Démonstration de l'impossibilité de la résolution algébrique des équations générales qui passent le quatrième degré,” 1824.
- [14] E. Galois, “Mémoire sur les conditions de résolubilité des équations par radicaux,” 1830s.
- [15] S. Lang, *Algebra*, Springer, New York.
- [16] D. S. Dummit and R. M. Foote, *Abstract Algebra*, Wiley, New York.
- [17] P. A. Samuelson, *Foundations of Economic Analysis*, Harvard University Press, Cambridge, 1947.
- [18] K. J. Arrow and G. Debreu, “Existence of an Equilibrium for a Competitive Economy,” *Econometrica*, Vol. 22, No. 3, 1954, pp. 265–290.
- [19] D. Fudenberg and J. Tirole, *Game Theory*, MIT Press, Cambridge, 1991.
- [20] M. J. Osborne and A. Rubinstein, *A Course in Game Theory*, MIT Press, Cambridge, 1994.
- [21] F. S. Mishkin, *The Economics of Money, Banking, and Financial Markets*, Pearson.
- [22] F. Modigliani and M. H. Miller, “The Cost of Capital, Corporation Finance and the Theory of Investment,” *American Economic Review*, Vol. 48, No. 3, 1958, pp. 261–297.
- [23] H. Markowitz, “Portfolio Selection,” *Journal of Finance*, Vol. 7, No. 1, 1952, pp. 77–91.
- [24] R. C. Merton, “Theory of Rational Option Pricing,” *Bell Journal of Economics and Management Science*, Vol. 4, No. 1, 1973, pp. 141–183.
- [25] T. C. Schelling, *The Strategy of Conflict*, Harvard University Press, Cambridge, 1960.
- [26] R. Axelrod, *The Evolution of Cooperation*, Basic Books, New York, 1984.

Glossary

Accounting Identity

The fundamental balance-sheet relation $A = L + E$, where A denotes assets, L liabilities, and E shareholders' equity.

Algebraic Closure

A mathematical constraint that ensures the components of a representation are mutually consistent.

Algebraic Balance Sheet

The term used in this paper for the cubic closure representation of the three-rate financial system. It is an analogy, not a literal accounting statement.

Central-Bank Rate

The policy interest rate associated with one of the three national central banks.

Elementary Symmetric Polynomial

A polynomial invariant under permutations of its arguments. For three variables, the first three are e_1 , e_2 , and e_3 .

Financial Account A

The aggregate rate $r_A = r_1 + r_2 + r_3$.

Financial Account B

The pairwise interaction account $r_B = r_1r_2 + r_2r_3 + r_3r_1$.

Financial Account C

The three-way interaction account $r_C = r_1r_2r_3$.

Cubic Financial Closure

The equation $r^3 - r_Ar^2 + r_Br - r_C = 0$ whose roots are the three central-bank rates.

Vieta's Relations

Relations connecting the coefficients of a polynomial to symmetric functions of its roots.

Discriminant

An algebraic quantity determining whether a polynomial has repeated roots and, for the cubic, characterizing the qualitative structure of its real roots.

Vandermonde Determinant

For three rates, $(r_1 - r_2)(r_1 - r_3)(r_2 - r_3)$, which vanishes exactly when two rates coincide.

Symmetry Breaking

The transition from equal or equivalent financial states to differentiated states.

Root Assignment

The institutional procedure by which the unordered roots of the common cubic are associated with particular nations.

Three-Nation Financial Closure

The property that a financial system with three genuine rates possesses a complete third-order elementary symmetric representation.

Financial Equilibrium

A state in which monetary-policy choices and economic conditions are mutually consistent with the strategic incentives of the participating institutions.

Political-Financial Equilibrium

A state in which political outcomes and financial conditions jointly satisfy their respective equilibrium mappings.

Financial Externality

An effect of one nation's financial policy or state on another nation's welfare or economic conditions.

Characteristic Polynomial

A polynomial whose roots represent characteristic quantities of an underlying system. Here it is $r^3 - r_A r^2 + r_B r - r_C$.

Symmetric Invariant

A quantity unchanged by permutation of the underlying national rates.

Nondegenerate Architecture

An architecture in which the three financial rates are genuine and the third-order invariant is not artificially eliminated.

Political-Financial Feedback

The process through which political choices affect financial conditions and financial conditions affect subsequent political choices.

Multilayer Network

A network containing several simultaneous relationship layers, such as finance, trade, diplomacy, military relations, and technology.

The End